usability evaluation in industry

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Taylor & Francis
Publishers since 1798

UK Taylor & Francis Ltd, 1 Gunpowder Square, London, EC4A 3DE
USA Taylor & Francis Inc., 1900 Frost Road, Suite 101, Bristol, PA 19007

British Library Cataloguing in Publication Data

A catalogue record for this book is available from the British Library.

ISBN 0 7484 0314 0 (cased)
ISBN 0 7484 0460 0 (paperback)

Library of Congress Cataloguing in Publication data are available

Cover design by Hans Jacobs and Robin Uleman (Philips Corporate Design).

Typeset in Times 10/12pt by MHL Typesetting Ltd, Coventry.

Printed in Great Britain by T.J. Press (Padstow) Ltd.

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Pat Jordan is a human factors specialist with Philips Corporate Design, where he is the leader of two large research projects. These investigate pleasure in product use and usability evaluation methods. Previously he was employed by the University of Glasgow in the Department of Psychology – first as a research assistant and then as a lecturer. He has been published 30 times in academic and technical journals, and books of conference proceedings. He presented and co-wrote the video *User Interface Performance Measurement* and co-authored an instructional CD-ROM on usability to be released in 1996. Dr Jordan is a regular speaker at international conferences and was General Chair of the seminar 'Usability Evaluation in Industry' on which this book is based.

Bruce Thomas studied psychology and worked briefly for a consultancy in the UK before moving to TüV Rheinland in Germany. There he worked on research, development and consultancy projects, including workplaces in power stations, hotel and restaurant kitchens and electronic equipment for cars. He spent a year on secondment to the University of Bremen investigating workplaces on ships. He has worked at Philips Corporate Design since 1990, where he is primarily responsible for coordinating human factors support for the development of communications products.

Bernard A. Weerdmeester graduated from the Agricultural University in Wageningen in 1982. After graduation he joined the Dutch PTT, where he carried out research on human factors in telecommunications. In 1986 he went to the University of Twente where he worked as a researcher and teacher of ergonomics. He returned to the PTT in 1989 and was head of the user interface group. With two former PTT colleagues he started up 'Raakvlak', a consultancy bureau on user interfaces. After a year, however, he left Raakvlak and started 'Usable', a bureau on the same market. Since 1987 Bernard Weerdmeester has been president of the

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Ian L. McClelland first trained as a mechanical engineer and then moved to Loughborough University of Technology in 1970 where he completed a post-graduate course in ergonomics. In 1972 he received his master's degree and joined the Institute for Consumer Ergonomics where he undertook a wide variety of research and consultancy work for government, business and industry. In 1986 he joined Philips Corporate Design as manager of the Applied Ergonomics Group. His interests include the integration of usability engineering principles into the process of user interface design, the involvement of users as 'partners' in interaction design, and evaluation methodology.

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Preface

As consumers become more and more sophisticated in terms of what they demand of products, so the role of the human factors specialist is becoming ever more challenging and demanding. Whilst users may once have seen ease of use as a bonus, in the case of many products it is now often taken for granted. In order to offer more, those involved with designing for usability now have to look beyond ease of use to consider issues such as the emotional and hedonic benefits associated with product use.

Philips Corporate Design is committed to excellence in design for usability, and human factors input is incorporated throughout the design process. We see activities such as the seminar 'Usability Evaluation in Industry', which brought together 25 of Europe's leading human factors practitioners and researchers, as an important forum for the exchange of ideas and views amongst those at the top of the profession. These professionals had the opportunity to learn from each other and to take the benefits of these shared ideas and approaches back to their own companies or research establishments. Philips Corporate Design was therefore happy to support the two-day event, both financially and through the commitment of the staff involved in the organization and running of the event.

The chapters in this book give an overview of the material presented at the seminar as well as the discussion that ensued. It should prove essential reading to those involved in the practice of human factors and all those involved in research to support usability issues.

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Acknowledgements

This book is based on the international seminar 'Usability Evaluation in Industry' that was held in Eindhoven, The Netherlands, on 14 and 15 September 1994. Financial support for the seminar came largely from Philips Corporate Design. We are grateful to them for supporting the idea of the seminar and giving us the necessary funding to put the idea into practice.

We would also like to thank the Dutch Ergonomics Society (Nederlandse Vereniging voor Ergonomie) for giving their support through endorsement of the seminar.

Finally, of course, we must thank the delegates for the chapters that make up this book and for two days of useful and stimulating discussion. In addition to those who contributed to the book, Kate Bentley of PTT Research and Gerard van der Heiden of Rabobank attended the seminar and made valuable contributions through their presentations and discussion.

Pat Jordan
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CHAPTER ONE

Introduction

PATRICK W. JORDAN, IAN L. McCLELLAND and BRUCE THOMAS

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Issues relating to usability are increasingly coming to prominence. This is reflected in, for example, the ever-growing literature relating to usability issues. This includes academic journals such as *Behaviour and Information Technology* and *Human–Computer Interaction*, books such as Nielson's *Usability Engineering* and Wiklund's *Usability in Practice*, and a host of newspaper and magazine articles. Similarly, there are now many national and international conferences and other gatherings which deal with usability issues. These include Computer–Human Interaction (CHI), the Human Factors and Ergonomics Society Conference and, of course, the seminar 'Usability Evaluation in Industry', which formed the basis for this book. Perhaps, though, the most significant reflections of how seriously usability issues are taken are the increasing number of professional human factors specialists employed in industry and the ongoing research efforts of academia in this area.

As the products that we use at home and in our workplaces become ever more complex in terms of features and functionality, it becomes vital that those involved in the design of these products consider the needs and limitations of those who will be using them. If these are not taken into account, products that are created with the intention of delivering some benefit can end up being more trouble than they are worth. Users are becoming more sophisticated with respect to their expectations about product performance. These include their expectations about a product's usability. Users seem no longer willing to accept having to struggle with products as the price to pay in return for 'technical wizardry'. Whilst once usability may have been seen as a bonus, it is rapidly becoming an expectation, with users becoming disenchanted with products which do not support an adequate quality of use. In the end these products will also disenchant those who are manufacturing them as they will find that their customers start looking elsewhere. Usability is becoming established as an important issue with respect to the marketing and sales of products, thus it can have significant commercial implications.

Manufacturers are continually seeking new ways of trying to gain a competitive edge. In the current business environment, the opportunity to make competitive gains in 'traditional' ways – such as by high reliability or reduction in manufacturing costs – is being continually eroded. A major reason for this is the sophistication of manufacturing processes, the high quality and consistency of which make it difficult for any one manufacturer to gain a significant advantage over another in these respects. Exploring the advantages of user-centred design is seen by some manufacturers as a way of enhancing possibilities to gain competitive advantage.

Most often products which are difficult to use may result in annoyance and frustration – for example, with video cassette recorders (VCRs) that are difficult to program, computer applications that we can't understand, or central heating systems that switch themselves on or off when we least expect. However, usability can also be associated with more serious issues, in particular the level of risk to personal safety in using a product.

Consider, for example, the case of a driver using in-car navigation equipment. It may take him or her, say, ten seconds to interpret navigation advice presented by a system. During this time the attention of the driver will be diverted from what's happening on the road. This is likely to make it more difficult to deal with anything unexpected that may happen – for example, a car stopping ahead or a child running out into the road. Similarly, with emergency products, such as fire extinguishers, it is vital that any member of the public can pick up the product and use it at the first time of trying. Again, this means that such products have to be carefully matched to users' expectations.

The importance of designing usable products is now receiving increasing attention from industry. Usability is coming to be seen as part of industry's responsibility to its customers, and, of course, it is also becoming important in terms of selling products. More and more products are advertised as being 'user-friendly' or 'ergonomically designed' and where once potential customers saw usability as a bonus it is increasingly becoming an expectation.

So how do these industry-based human factors specialists go about ensuring that the products that their organizations produce are usable? This book is a collection of chapters from many specialists working in industry as well as academics who work closely with industry. The chapters describe a number of techniques used in the evaluation of products – techniques designed to give an overview of the suitability of the product for the user and to feed back information that can be used to improve the quality of the design. The authors also relate their experiences of applying their techniques in an industrial setting, where they are working under pressures and constraints which may be considerably sharper than and certainly different from those experienced in academia where the majority of the techniques were originally developed. The contributors to this volume met together for a two-day seminar in Eindhoven, The Netherlands, in September 1994, from which the chapters included in this book emerged. A number of issues also emerged during the course of the seminar discussions. These issues are addressed in Chapter 26, in which overall conclusions are drawn on the basis of what emerged from the discussions during the seminar.

This book should be of benefit to all those involved in product design and the evaluation of products for usability. Through their experiences the authors have gained clear ideas of what does and doesn't work under industrial constraints, and their chapters should be a valuable source of advice and support to those working under the same pressures. This can, perhaps, be seen as a contrast with more academic writings – such as those contained in journals or conference proceedings – where the emphasis may be more on scientific validity and procedural rigour, rather than on the pragmatics of achieving a useful result as quickly and cheaply as possible.

This book should also prove valuable to academia. Research aimed at the support of usability evaluation is increasingly being carried out at universities and other research establishments throughout the industrialized world. Often, however, there may be a gap between what academia provides and what industry finds helpful. A contributory factor to this may well be a lack of understanding of industrial practices and pressures. Again, material relating to this may be difficult to get hold of in journals and conference proceedings. Certainly, many academics will work with individual companies and may gain an understanding of how some industry-based ergonomists work. However, the experiences related in this book should give a broader overview, demonstrating the practice of industry-based usability evaluation as a whole.

PART ONE

Elements of Usability Evaluation

CHAPTER TWO

A combined effort in the standardization of user interface testing

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INTRODUCTION

Philips, being a multinational consumer electronics company, needs to conduct studies of the usability of its products with users in a variety of countries, settings, etc. This means that investigations, or parts of them, must be conducted by local agencies. One of the problems at the moment is that for individual projects (depending on, for example, the market research agency used), the approaches, analysis of results, reporting, etc. are often different. This makes it impossible to compare the results of tests done on interfaces for different TV sets, for example. Since there is a drive for continuous improvement within Philips, we need to be able to use previously developed interfaces as benchmarks.

This chapter presents some of the results of a joint project between the market research department of one of Philips' Business Groups (BG), Television (TV), and the Applied Ergonomics group of Philips Corporate Design (PCD). Since this has been a joint project, some of the techniques described are traditional human factors techniques, whereas others are derived from market testing.

The goal of this project has been to develop a standardized approach to the evaluation of user interfaces for BG, taking a very pragmatic approach. The end product was to be a document describing which techniques to use when, what kind of questionnaire to use, etc. This document will serve not only as an internal checklist and selection instrument, but also as a set of 'rules' to be followed when an outside contractor is employed. The issues to be addressed were defined at the start of the project:

- purpose of test,
- specification of subjects,
- design of user/usability tests,
- measures to be taken,
- questionnaires used,
- interview questions,
- tasks to be executed,
- presentation of results.

In the following, excerpts from the original document resulting from this project (which is still in progress) are presented and briefly discussed. To help structure the approach, the methods and techniques are linked to the product creation process (PCP) of this specific BG. The PCP, which is predetermined by BG TV, consists of the following main phases:

- Know-how phase: Aims at gathering, generating and applying know-how in several fields.
- Concept phase: Encompasses carrying out work for future product generations.
- After product range start (PRS): The phase leading up to the specification of a rough product range.
- After commercial release (CR): The product is released for delivery and distribution on the market.

The term 'test' is used in this chapter to mean any involvement of users in getting feedback on the user interface of a television. The terminology used here may not always be in line with 'standards' used within the human factors community, but corresponds to that used throughout Philips.

PURPOSES OF USER INTERFACE TESTS

Generic purposes

The purpose of a test will influence the methods and techniques to be used, and what is to be measured. Hence it is of great importance to define the purpose of the test before starting. Some fundamental and generic questions that can/must be addressed regarding the purpose are for example:

Are we trying to find out what people want with regard to user interfaces?
A starting point can be to determine the customer's needs and wishes, which can be translated into a user requirements specification.

Are we trying to find out what people accept with regard to user interfaces?
Taking a given system, the goal of a test might be to determine the acceptance of such a system by our customers.

Are we trying to find out what people are able to use?
Another goal might be to test to what extent consumers are able to use, independent of the aesthetic qualities, a system they are confronted with, e.g. whether there are problems with particular interactions or interaction mechanisms.

Specific purposes related to the PCP

Although the purpose of user interface (UI) tests has to be fine tuned with each new test for each new project, some more specific purposes can be distinguished, based on the phase in the PCP in which the test will be conducted. Some examples for each phase are:

Know-how phase
Testing to generate new ideas for concepts
Testing to select the first basic ideas and concepts
Inventory of existing problems
Testing to answer specific questions

Concept phase
Testing to answer specific questions
Testing to determine whether acceptance criteria have been met
Testing to select ideas and concepts
Testing to identify usability problems

After PRS
Testing the final concept in detail, and verification of implementation

After CR
Inventory of problems and remarks
Confirmation of usability criteria
Detailing of problems

SPECIFICATION OF THE PARTICIPANTS

Before each test, a detailed specification of the participants has to be made based on the product brief (with reference to the target group(s), etc.). The idea is to make use of a checklist in which the main characteristics of participants are listed. Before starting a test those responsible have to go through this list to specify the participants needed. In principle, much of this information can be found in a User Requirement Specification. Examples of the checklist items for participants are: